

Hoja de Trabajo 1

1)

$$a) \frac{13\pi}{5} = \frac{13\pi}{5} \left(\frac{180^\circ}{\pi} \right) = \underline{468^\circ}$$

Angulos coterminales

$$\frac{3\pi}{5} \quad -\frac{7\pi}{5}$$

Angulo ref.

$$\frac{3}{5} = 0.6 \rightarrow \underline{\text{II}}$$

$$\pi - \frac{3\pi}{5} = \underline{\frac{2\pi}{5}}$$

$$b) \frac{37\pi}{9} = \frac{37\pi}{9} \left(\frac{180^\circ}{\pi} \right) = \underline{740^\circ}$$

Angulos coterminales

$$\frac{\pi}{9} \quad -\frac{17\pi}{9}$$

Angulo ref.

$$\frac{1}{9} = 0.11 \rightarrow \underline{\text{I}} \quad \frac{\pi}{9}$$

$$c) \frac{39\pi}{8} = \frac{39\pi}{8} \left(\frac{180^\circ}{\pi} \right) = \underline{877.5^\circ}$$

Angulos coterminales

$$\frac{7\pi}{8} \quad -\frac{9\pi}{8}$$

Angulo ref.

$$\frac{7}{8} = 0.88 \rightarrow \underline{\text{II}}$$

$$\pi - \frac{7\pi}{8} = \underline{\frac{\pi}{8}}$$

$$d) \frac{29\pi}{4} = \frac{29\pi}{4} \left(\frac{180^\circ}{\pi} \right) = \underline{1305^\circ}$$

Angulos coterminales

$$\frac{5\pi}{4} \quad \quad \quad -\frac{3\pi}{4}$$

Angulo ref.

$$\frac{5}{4} = 1.25 \rightarrow \text{III}$$

$$\frac{5\pi}{4} - \pi = \frac{\pi}{4}$$

$$e) \frac{123\pi}{12} = \frac{123\pi}{12} \left(\frac{180^\circ}{\pi} \right) = \underline{1845^\circ}$$

Angulos coterminales

$$\frac{\pi}{4} \quad \quad \quad -\frac{7\pi}{4}$$

Angulo ref.

$$\frac{1}{4} = 0.25 \rightarrow \text{I} \quad \quad \quad \frac{\pi}{4}$$

2) Obtenga el area de un sector que tiene un arco de 6 pies de longitud en un círculo de 4 pies de diámetro.

$$S = r\theta$$

$$A = \frac{1}{2}(2)^2 3$$

$$6 = 2\theta$$

$$6/2 = \theta$$

$$\theta = 3$$

$$A = 6 \text{ pies}^2$$

3) 1) 20 rpm

$$\omega_{\text{rad}} = 20 \times \frac{2\pi}{1 \text{ rev}} = 40\pi \text{ rad/min} = \omega$$

$$V = r\omega = 10(40\pi \text{ rad/min}) = 400\pi \text{ /min}$$

$$20 = \omega = \frac{V}{r} = \omega = \frac{400\pi \text{ /min}}{20} = 20 \text{ rad/min}$$

$$\omega = \frac{V}{r}$$

$$\omega = \frac{400\pi \text{ /min}}{40}$$

$$\omega = 10\pi \text{ rad/min}$$

$$\text{rpm} = \frac{\pi \text{ rad}}{\text{min}} \times \frac{1 \text{ rev}}{2\pi} = \underline{5 \text{ rpm} = 4}$$

4)

$$0. \cos \alpha = \frac{4}{11}$$

$$\cos \alpha = \frac{1}{\sec \alpha} = \frac{4}{11} \quad \sec \alpha = \frac{11}{4}$$

$$\tan^2 \alpha = \left(\frac{11}{4}\right)^2 - 1$$

$$\tan \alpha = \frac{\sqrt{105}}{\sqrt{16}}$$

$$\tan \alpha = \frac{\sqrt{105}}{4}$$

$$\tan \alpha = \frac{1}{\cot \alpha} = \frac{\sqrt{105}}{4} \quad \cot \alpha = \frac{4}{\sqrt{105}}$$

$$\tan \alpha = \frac{\sec \alpha}{\cos \alpha}$$

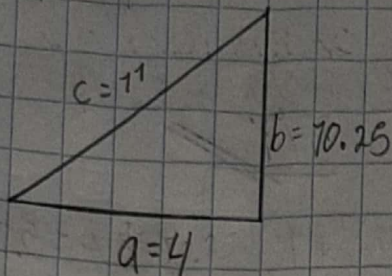
$$\frac{\sqrt{105}}{4} = \frac{\sec \alpha}{\frac{4}{11}}$$

$$\sec \alpha = \frac{\sqrt{105}}{11}$$

$$\sec \alpha = \frac{1}{\cos \alpha} = \frac{\sqrt{105}}{11}$$

$$\csc \alpha = \frac{11}{\sqrt{105}}$$

$$\cos \alpha = \frac{4}{11}$$



$$\begin{aligned} 4^2 + b^2 &= 11^2 \\ b^2 &= 105 \\ b &= \sqrt{105} = 10.25 \end{aligned}$$

$$\sin \alpha = \frac{\sqrt{7}}{11}$$

$$\sin \alpha = \frac{1}{\csc \alpha} = \frac{\sqrt{7}}{11}$$

$$\csc \alpha = \frac{11\sqrt{7}}{7} \approx 7.18$$

$$\sin^2 \alpha + \cos^2 \alpha = 1 \quad \cos^2 \alpha = 1 - \left(\frac{\sqrt{7}}{11}\right)^2$$

$$\cos \alpha = \frac{354}{361} \approx 0.98$$

$$\tan \alpha = \frac{\sin \alpha}{\cos \alpha}$$

$$\tan \alpha = \frac{\frac{\sqrt{7}}{11}}{\frac{354}{361}}$$

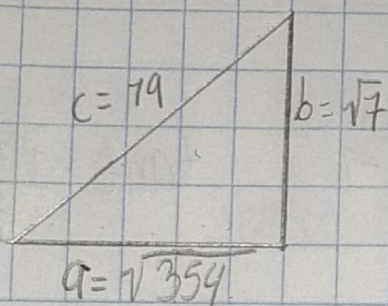
$$\tan \alpha = \frac{11\sqrt{7}}{354} \approx 0.14$$

$$\tan \alpha = \frac{1}{\cot \alpha} = \frac{11\sqrt{7}}{354}$$

$$\cot \alpha = \frac{354}{11\sqrt{7}} \approx 7.01$$

$$\cos \alpha = \frac{1}{\sec \alpha} = \frac{354}{361}$$

$$\sec \alpha = \frac{361}{354} \approx 2.24$$



$$\begin{aligned} a^2 + \sqrt{7}^2 &= 11^2 \\ a^2 &= 361 - 7 \\ a &= \sqrt{354} \end{aligned}$$

$$C. \tan \alpha = \frac{11}{23}$$

$$\tan \alpha = \frac{1}{\cot \alpha} = \frac{11}{23}$$

$$\cot \alpha = \frac{23}{11} \approx 2.09$$

$$\sec^2 \alpha = \tan^2 \alpha + 1$$

$$\sec^2 \alpha = \left(\frac{11}{23}\right)^2 + 1$$

$$\sec \alpha = \frac{5\sqrt{26}}{23} \approx 1.11$$

$$\sec \alpha = \frac{1}{\cos \alpha} = \frac{5\sqrt{26}}{23}$$

$$\cos \alpha = \frac{23}{5\sqrt{26}} \approx 0.90$$

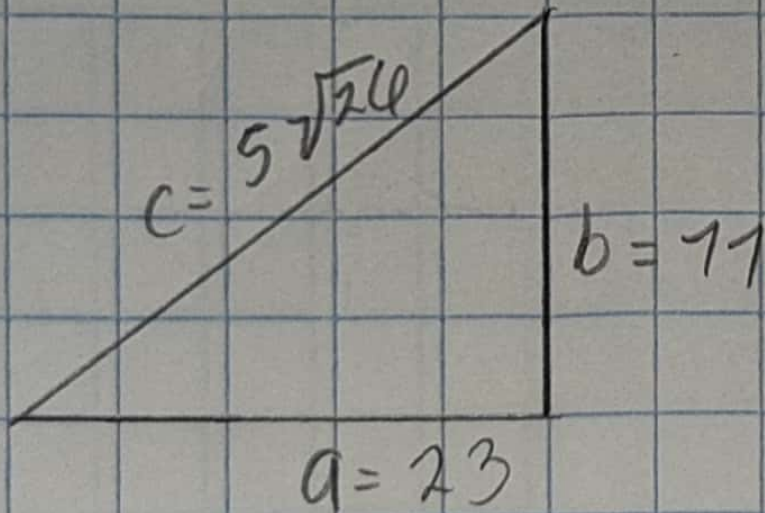
$$\sin^2 \alpha + \cos^2 \alpha = 1$$

$$\sin^2 \alpha = 1 - \left(\frac{23}{5\sqrt{26}}\right)^2$$

$$\sin \alpha = \frac{11\sqrt{26}}{730} \approx 0.43$$

$$\sin \alpha = \frac{1}{\csc \alpha} = \frac{11\sqrt{26}}{730}$$

$$\csc \alpha = \frac{5\sqrt{26}}{11} \approx 2.32$$



$$\begin{aligned} 23^2 + 11^2 &= c^2 \\ 529 + 121 &= c^2 \\ \sqrt{650} &= c \\ 5\sqrt{24} &= c \end{aligned}$$

5)

	Coseno	seno	tangente	cotangente	secante	cosecante
1. $\sin \alpha \geq 0$	$\frac{4}{7}$	$\frac{\sqrt{33}}{7}$	$\frac{\sqrt{33}}{4}$	$\frac{4\sqrt{33}}{33}$	$\frac{7}{4}$	$\frac{7\sqrt{33}}{33}$
2. $\tan \leq 0$	$\frac{359}{79}$	$\frac{\sqrt{2}}{79}$	$\frac{\sqrt{718}}{359}$	$\frac{359}{718}$	$\frac{79\sqrt{359}}{359}$	$\frac{79\sqrt{2}}{2}$
3. $\sec \geq 0$	$\frac{3}{\sqrt{70}}$	$\frac{\sqrt{70}}{70}$	$-\frac{1}{3}$	-3	$\frac{\sqrt{70}}{3}$	$\sqrt{70}$

$$1. \cos \theta = \frac{1}{\sec \theta} = \frac{1}{\frac{7}{4}} = \frac{4}{7} \quad \sec \theta = \frac{7}{4}$$

$$\sec^2 \theta = \tan^2 \theta + 1 \quad \tan^2 \theta = \left(\frac{7}{4}\right)^2 - 1 \quad \tan \theta = \frac{\sqrt{33}}{4}$$

$$\tan \theta = \frac{1}{\cot \theta} = \frac{\sqrt{33}}{4}$$

$$\cot \theta = \frac{4\sqrt{33}}{33}$$

$$\csc^2 \theta = \cot^2 \theta + 1$$

$$\csc^2 \theta = \left(\frac{4\sqrt{33}}{33}\right)^2 + 1$$

$$\csc \theta = \frac{7\sqrt{33}}{33}$$

$$\csc \theta = \frac{1}{\sin \theta} = \frac{7\sqrt{33}}{33}$$

$$\sin \theta = \frac{\sqrt{33}}{7}$$

$$2. \sin \theta = \frac{1}{\csc \theta} = \frac{\sqrt{2}}{72}$$

$$\csc \theta = \frac{72\sqrt{2}}{2}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\cos^2 \theta = 1 - \left(\frac{\sqrt{2}}{72}\right)^2$$

$$\cos \theta = \frac{\sqrt{359}}{19}$$

$$\cos \theta = \frac{1}{\sec \theta} = \frac{\sqrt{359}}{79}$$

$$\sec \theta = \frac{79\sqrt{359}}{359}$$

$$\sec^2 \theta = \tan^2 \theta + 1$$

$$\tan^2 \theta = \left(\frac{79\sqrt{359}}{359}\right)^2 - 1$$

$$\tan = \frac{\sqrt{718}}{359}$$

$$\tan \theta = \frac{1}{\cot \theta} = \frac{\sqrt{718}}{359}$$

$$\cot \theta = \frac{359}{\sqrt{718}}$$

3.

$$\tan \theta = \frac{1}{\cot \theta} = -\frac{1}{3}$$

$$\cot = -3$$

$$\sec^2 \theta = \tan^2 \theta + 1$$

$$\sec^2 \theta = \left(-\frac{1}{3}\right)^2 + 1$$

$$\sec \theta = \frac{\sqrt{10}}{3}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - \left(\frac{3}{\sqrt{10}}\right)^2$$

$$\sin \theta = \frac{\sqrt{10}}{70}$$

$$6) \csc(x) \sin(x) \tan(x) + \sec(x) \cot(x), \sin(x)$$

$$\left(\frac{1}{\cancel{\sin(x)}} \sin(x) \right) \left(\frac{\sin(x)}{\cos(x)} \right) + \frac{1}{\cancel{\cos(x)}} \left(\frac{\cos(x)}{\sin(x)} \right)$$

$$\frac{\sin(x)}{\cos(x)} + \frac{1}{\sin(x)} = \frac{\sin^2(x) + \cos(x)}{\cos(x) \sin(x)}$$

$$\frac{\sin^2(x) + \sqrt{1 - \sin(x)}}{\sqrt{1 - \sin(x)} \sin(x)}$$

$$\cos^2(x) = 1 - \sin^2(x)$$

$$\cos(x) = \sqrt{1 - \sin(x)}$$

$$7) \frac{\csc x - \cot x}{\sec x - 1} = \cot x$$

$$\frac{\frac{1}{\sin x} - \frac{\cos x}{\sin x}}{\frac{1}{\cos x} - 1} = \cot x$$

$$\frac{\frac{1 - \cos x}{\sin x}}{\frac{1 - \cos x}{\cos x}} = \cot x$$

$$\frac{\cos x}{\sin x} = \cot x$$

$$\cot x = \cot x$$